Couchbase

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Couchbase Enterprise Technical Overview

Design Principles and Key Differentiators

March 2026

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Agenda

01/ Couchbase Enterprise Overview

02/ Elastic Architecture

03/ Built-in HA

04/ Performance at Scale

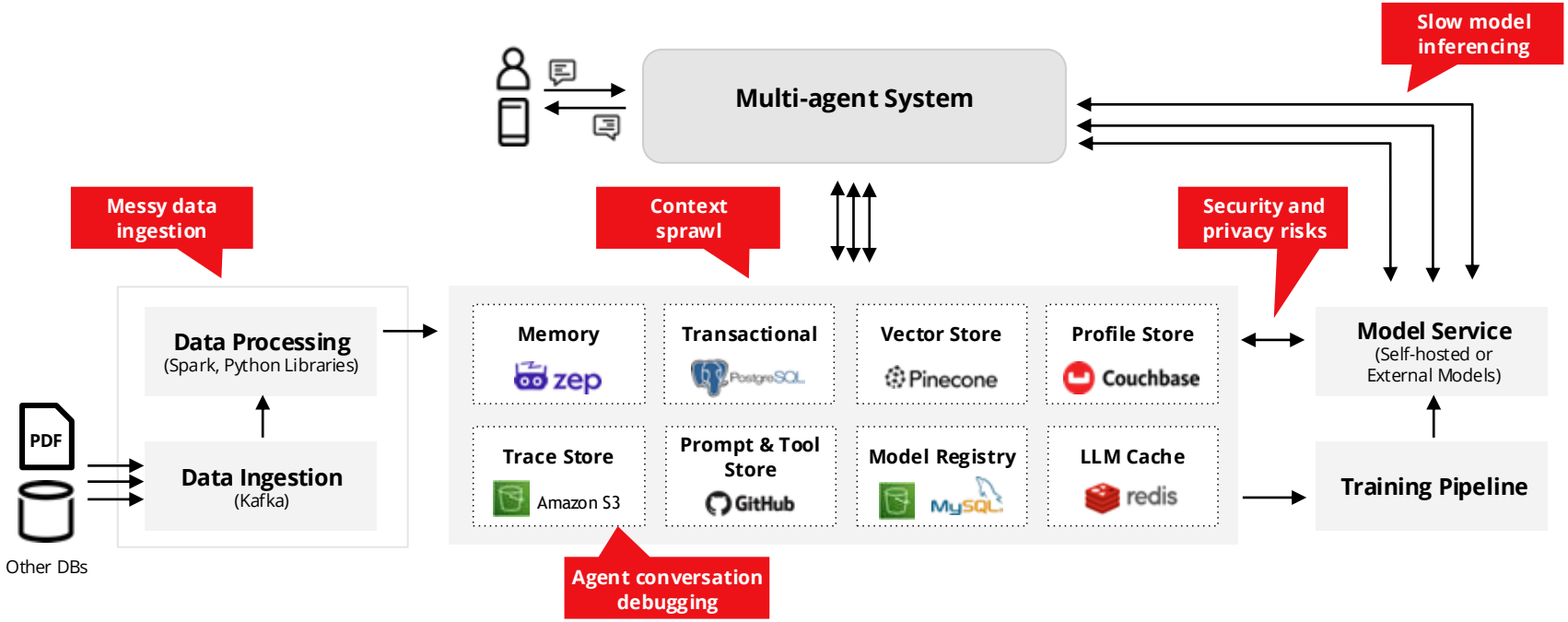
05/ Cloud to Edge Deployment

06/ Develop and Run Easily

07/ Key Takeaways

Couchbase Enterprise Overview

Yesterday's Data Architecture Design is Not Ready for AI



Fragmented data + manual integrations = Big challenges: Architecture, latency, management, trust & cost

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Couchbase's Developer Data Platform

A single unified data layer delivering speed, flexibility, scale and reliability for a variety of workloads and use cases.

User Applications and Agents

Agentic AI *RAG Chatbots* *Predictive ML* *Product Catalog* *Operational Analytics/BI*
Field Service Apps *IoT* *Customer 360* *Profiles & Personalization* *Caching & Session Management*

Other Data Interactions

Connectors

Semistructured

Unstructured

Multi Source

Data Access (SDKs, SQL+, MCP, Data API, Natural Language)

JSON

Operational	**Mobile Edge**	**Analytical**	**AI**
Key-value cache & persistence	*Sync & offline mobile persistence*	*Columnar persistence*	*Billion scale vector index*

Enterprise-grade Security (SOC 2, HIPAA, GDPR, PCI DSS, ISO 27001)

AI Integrations

Model Platforms

MCP Registries

Frameworks & Tools

Security & Observability

Data Catalog & Governance

ENTERPRISE (Customer Managed)

Public Cloud | Cloud Edge | On Prem

CAPELLA (As-a-Service)

Public Cloud

Enterprise Server | Innovation Over Time

Data Access Services

Performance Foundation

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Design Principles for Couchbase Data Platform

Elastic Architecture	Built-In High Availability	Performance at Scale	Cloud to Edge Deployment	Develop & Run Easily
Designed as a clustered and replicated distributed database from day 1	Data is automatically partitioned and replicated across nodes	Built-in and fully managed object cache for low latency at scale	Active-active XDCR for global deployments	Declarative query language on flexible document model; Programmatic SDKs
100% of customers rely upon clustering and rebalance	Built-in rack zone awareness with smart data placement	High performance storage engine for efficient persistence	Multi and hybrid-cloud deployments	Integrated platform powering emerging AI applications
Application behavior unchanged from 1 node on a laptop, to 3 nodes in UAT, to 20 nodes in production	Failure detection and auto-failover in seconds	Billion scale vector indexes w/ best-in-class performance	Cloud-to-edge with automatic data synchronization	All capabilities available across physical, virtualized, public cloud, container and fully managed service

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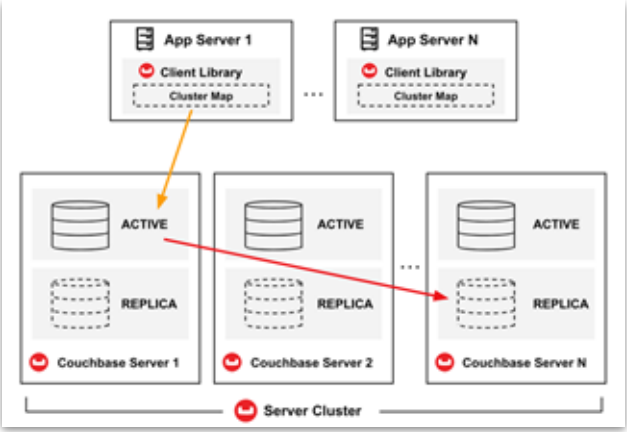
Elastic Architecture

Elastic Architecture | From Day 1



DAY 1

Designed as a clustered and replicated distributed database from **day 1**.

COUCH

COUCH is the acronym for “Cluster Of Unreliable Commodity Hardware”.

100%

100% of our features rely on the elastic architecture.

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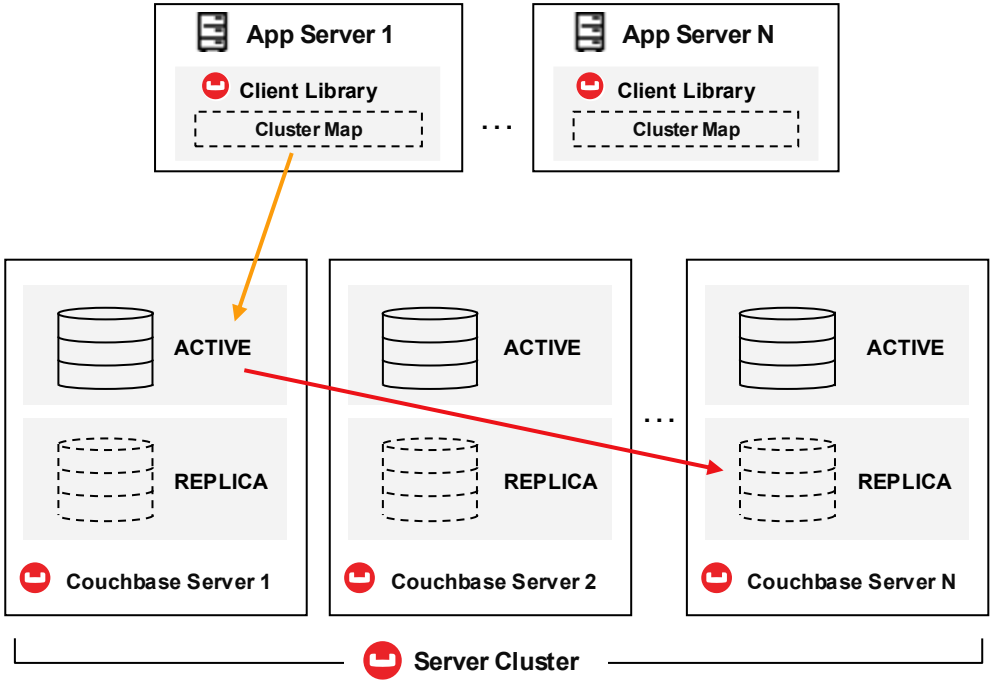
Elastic Architecture | Key Capabilities

- Scale on demand with **automatic** partitioning and rebalancing
- Build **always-available apps** with memory-to-memory replication and automatic failover
- **Simplify development** with topology-aware clients and direct communication
- **All features** work with automatic partitioning and replicas - **no change** and **no downtime** in applications as you **evolve** your data topology
- **CP database** (CAP theorem) with configurable consistency/durability level per request and support for ACID transactions



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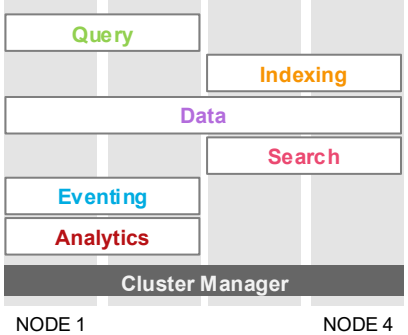
Elastic Architecture | Multi-Dimensional Scaling

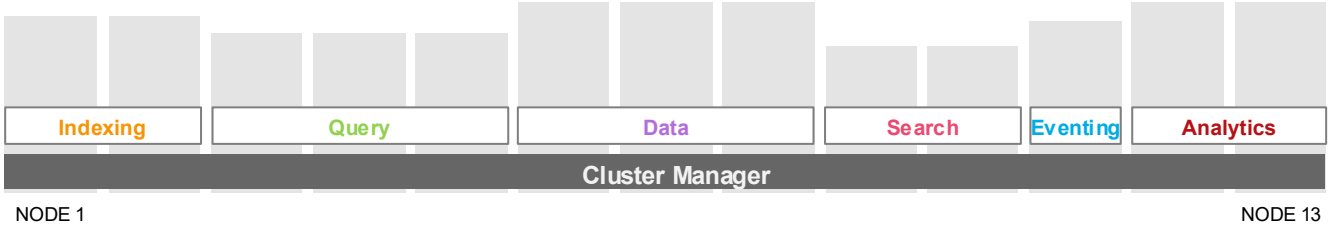
Sample Dev Setup

Sample QA Setup



- **Independent scaling** based on the needs of each service
- Ability to **co-locate or isolate** workloads based on application requirements
- Service **appropriate hardware**
- **No change** in application code

Sample Production Deployment



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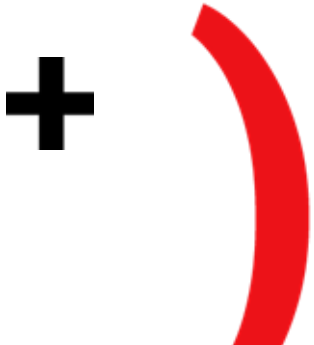
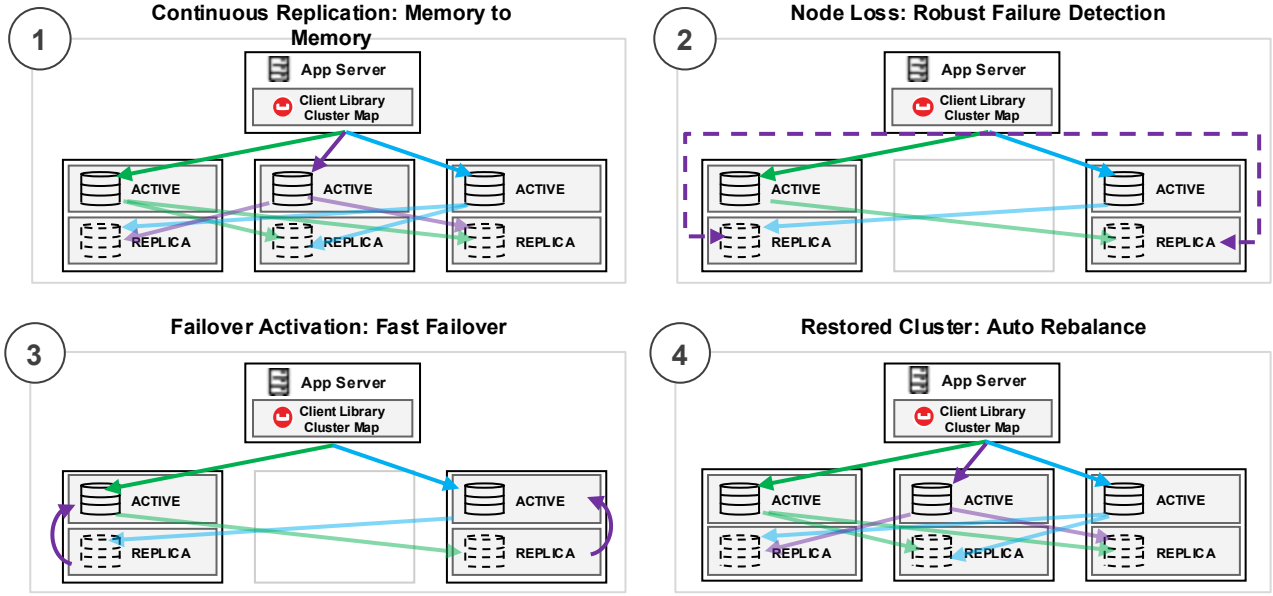
Built-In HA



Built-in HA | Always-On Reliability



- **Memory-memory** replication to reduce data loss
- Option to use durable writes
- **Fast and robust failure detection**
- Fast failover and automatic rerouting of app requests
- Data auto-rebalancing when the failed node is restored
- Smart placement of data and replicas to handle **rack** and **availability zone** failures

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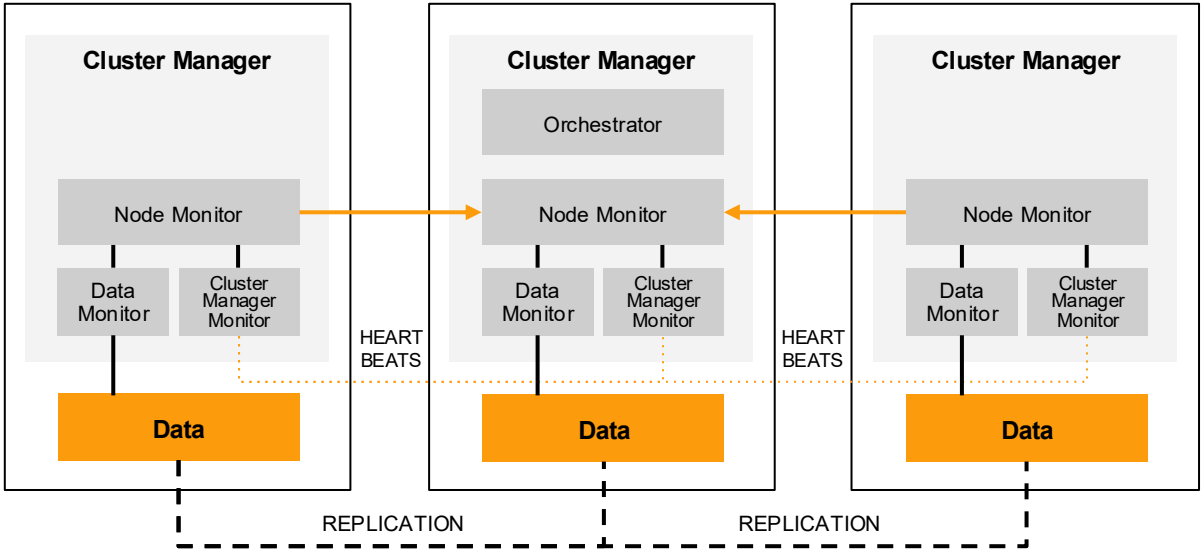
Built-in HA | Fast Failover with Robust Failure Detector

Fast failover in seconds (incl. automatic SDK traffic redirection) and robust failure detector to avoid false positives or missed failures



- **Cluster Manager Monitor** tracks health of cluster manager on every node using heartbeats
- **Data Monitor** checks inter-cluster replication traffic and health of data service locally and on other nodes
- **Node Monitor** consolidates health data and sends to Orchestrator node
- **Node Status Analyzer** on Orchestrator analyzes view of all monitors on all nodes to determine the health status of each node in the cluster

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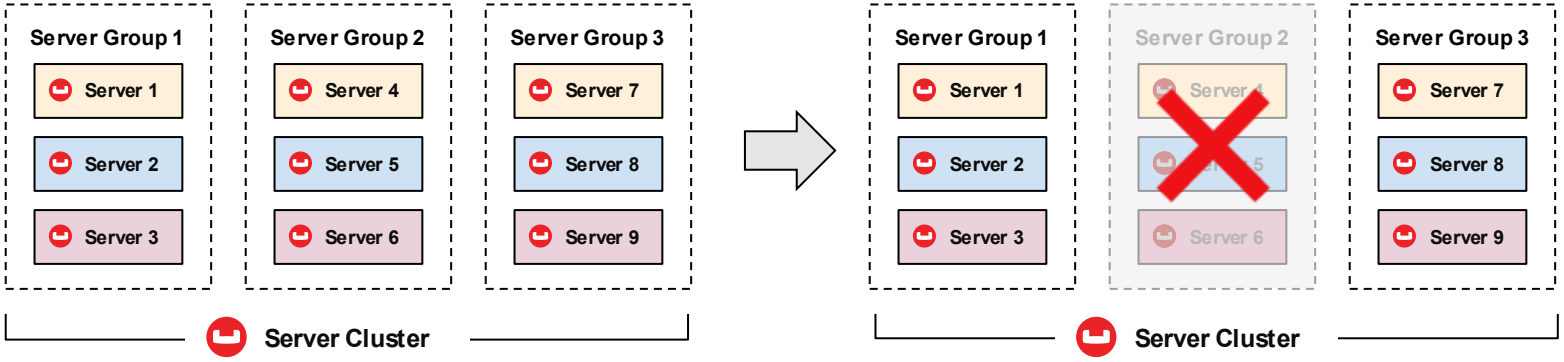
Built-in HA | Handle Rack and AZ Failures

- **Server Groups - Rack/Zone Awareness**
- Informs Couchbase as to the physical distribution of nodes in **racks** or **availability zones**
- Ensures that active and replica vBuckets are **distributed across groups**
- If an **entire rack fails** or an **entire cloud availability zone fails**, data is still available



Cluster has 2 replicas (3 copies) across 3 groups

If an entire group fails, data is still available

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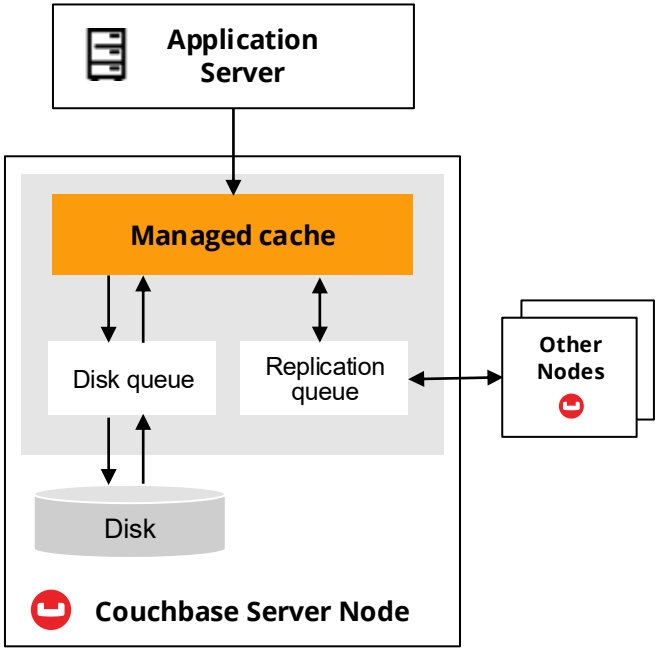
Performance at Scale

Managed Cache

Couchbase Server provides a fully integrated caching layer, which provides high-speed data-access

- Applications write and read data from the managed cache
- Couchbase Server automatically manages the cache
- Sub-millisecond latency for read and write operations



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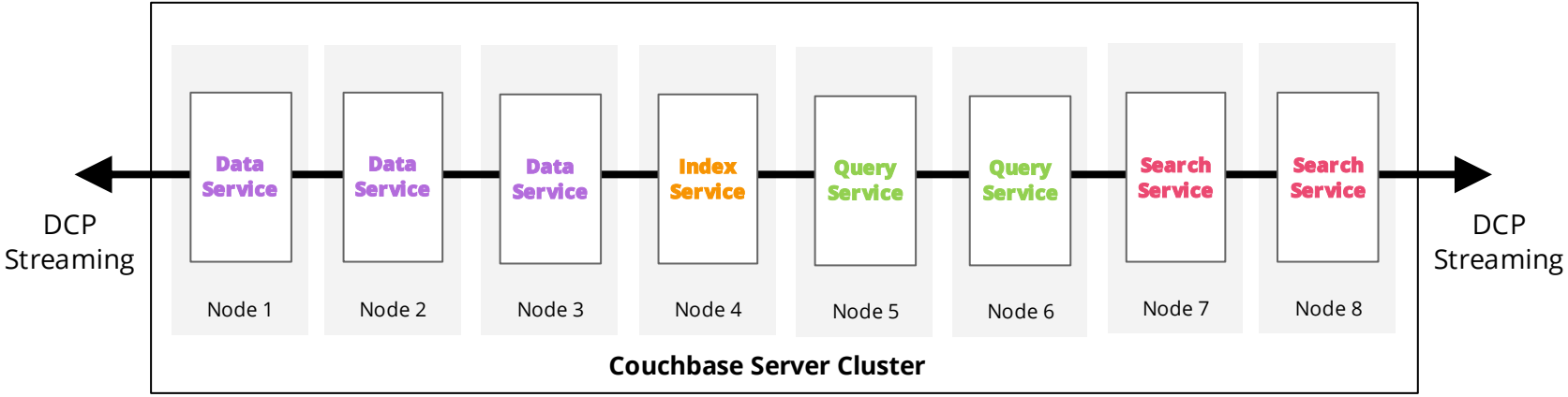
Memory-first Asynchronous Architecture

Data movement free from disk bottlenecks



In-memory streaming of updates to all components and XDCR

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Independent Scalability | Multi-Dimensional Scaling

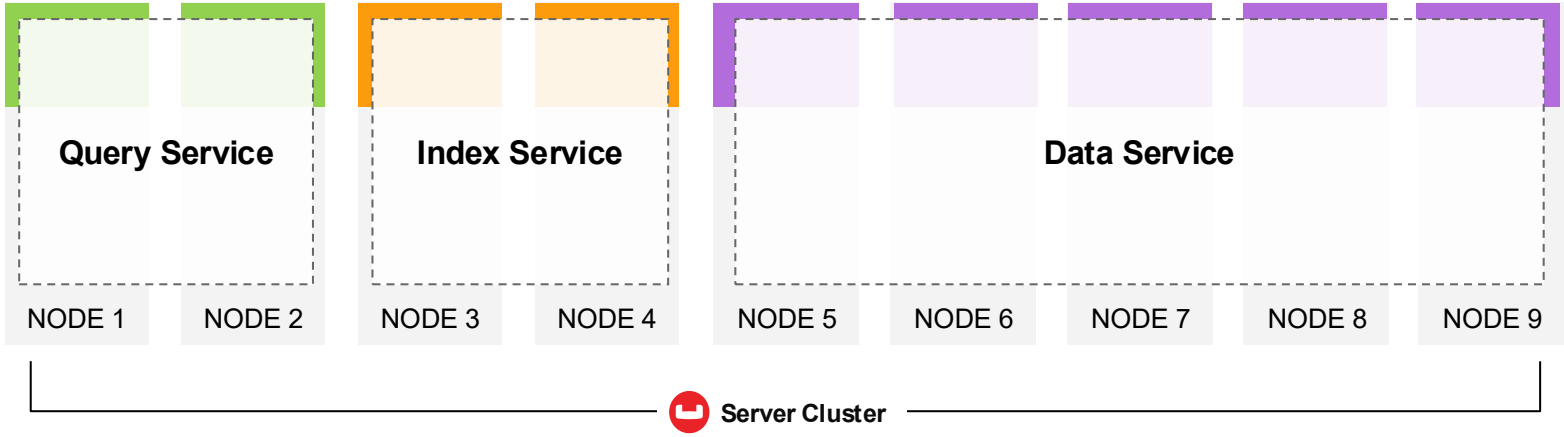
Independent Scalability for Best Computational Capacity – per Service

Heavier Indexing?

Scale up or out **Index Service** Nodes

More RAM for Data Processing?

Scale up or out **Data Service** Nodes



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Data Storage Engine | Magma

Couchbase on Magma is a high performance disk-oriented database for large volumes of data

High Data Density at Low TCO

< 1% memory to data ratio

- Store 10TB on a single node using < 100GB memory
- 100s of TB to PB scale clusters

High Disk Access Performance

Exceeds state-of-art alternatives (*)

- High sustained write throughput
- Better disk read latency

Scale Up with High-End Hardware

Scale with NVMe SSDs

- Scale better with higher disk bandwidth for write heavy workloads
- Lower read latency with NVMe drives

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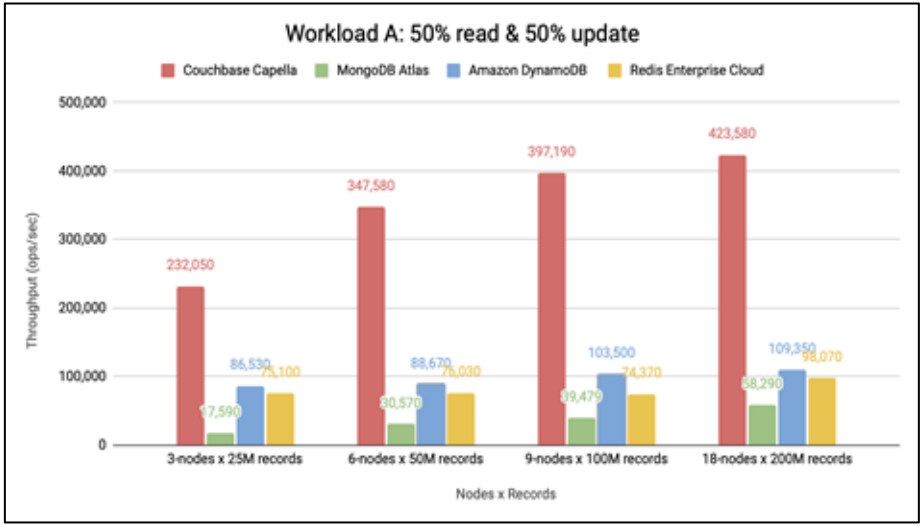
(*) See https://vldb.org/pvldb/vol15/p3496-lakshman.pdf

Performance Benchmark | YCSB

Couchbase excels at raw performance and at scaling



https://www.couchbase.com/content/capella/altoros-report-eval-nosql-dbaas

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3 Types of Vector Indexes: Flexibility Give Better Performance

Choosing the right vector index depends on your workload priorities

Performance — **Data volume** — **Accuracy** — **Cost efficiency** — **Search flexibility**

7.6 **Search Vector Index**

8.0 **Composite Vector Index**

8.0 **Hyperscale Vector Index**

Use: Hybrid Search
- Best for search scenarios which involves vector, full text, parameters, and geospatial locations

Use: Pre-filtered Search
- Best for advanced pre-filtering scenarios when developers control prompts

Use: RAG, Agents, & Recommendations
- Best-in-class Performance (Accuracy, Queries per second and low latency)
- Large corpus of vectors for any prompt

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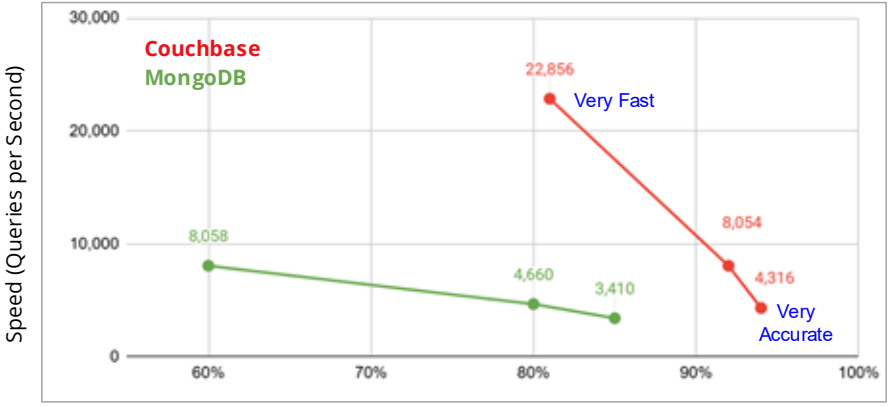
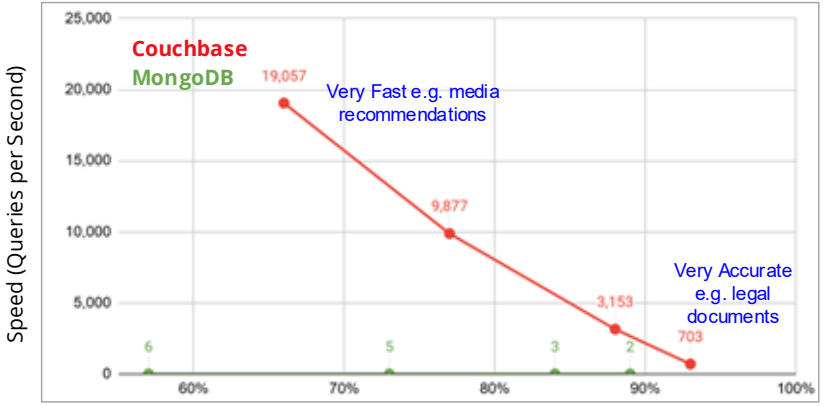
Results: Couchbase vs. MongoDB

VectorDBBench Standard on equivalent hardware

Speed (Queries per Second)
[100M Dataset, 768 dimensions]



Speed (Queries per Second)
[1B Dataset, 128 dimensions]



Couchbase delivers *higher* throughput, even with when more *accuracy* is needed

- **100M vectors:** Couchbase is 5X faster at same accuracy, and can deliver more accurate results
- **1B vectors:** MongoDB cannot deliver at this scale

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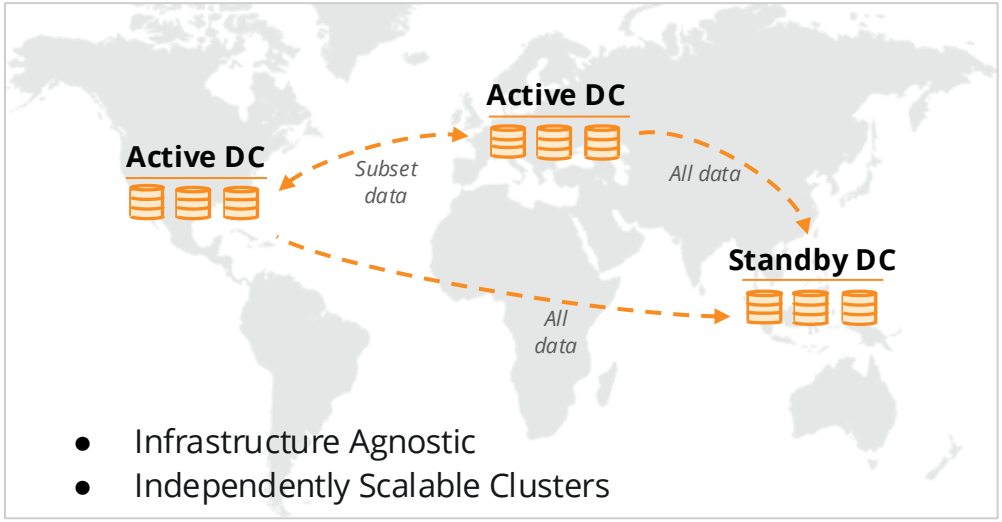
Data points represent focus intensity setting that improves accuracy

Cloud to Edge Deployment

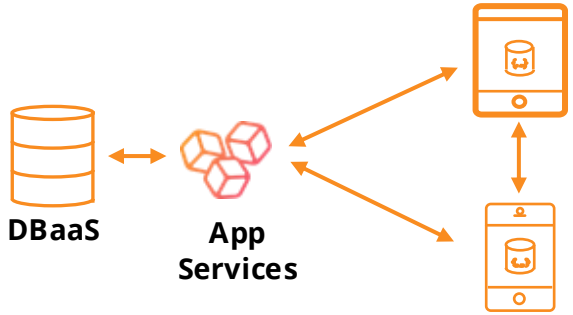
Global Cloud to Edge Deployments

Geo-Replication Service



- Infrastructure Agnostic
- Independently Scalable Clusters

Mobile Sync



- Low latency to the Edge
- Peer–to-peer syncing

Couchbase Managed	Customer-Managed
Couchbase CAPELLA; aws, Azure, Google Cloud — Public Cloud	aws, Azure, Google Cloud, RedHat, kubernetes — Public Cloud; verizon, aws Outposts — Cloud Edge; On Prem; Edge Devices

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Couchbase Mobile | Product Stack

Cloud to Edge

Couchbase Enterprise

Scalable, performant NoSQL data platform

Sync Gateway

Secure gateway for data sync and access control

Edge Server

Resource constrained Edge database server

Couchbase Lite

Full featured NoSQL embedded database

Deploy Fully Managed

Couchbase CAPELLA

aws

Deploy Self-Managed

Public Cloud

aws

Private Cloud

Cloud Edge

verizon

aws Outposts

On Prem

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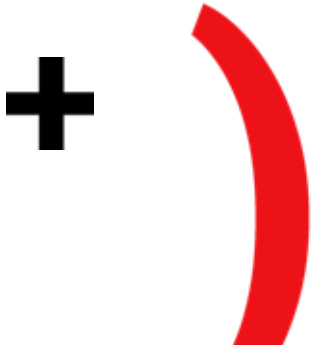
Develop & Run Easily



The Developer Data Platform for Critical Applications in Our AI World

Eliminate Multiple Databases and Caches
Simplify | Lower latency | Save

Proven Speed and Flexibility
In-memory architecture | Flexible JSON | Advanced indexing

Broad Data Capabilities
Vector search | Text search | Graph | Real-time analytics | Powerful SQL++

Advanced Vector Across our Products
On-prem | Cloud | Offline mobile

For Safe, Scalable Enterprise AI
Unstructured data | Automated vectorization | Model Service | AI Functions | Agent Catalog

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Flexibility of JSON | Relaxed Normalization

Store data in logical ways

- Denormalized single document
- Normalized with references
- Add new values when needed
- Support for binary values

Access Data in multiple ways

- Direct Key-Value
- SQL++ querying
- Full-Text Search
- Semantic Vector Search
- MPP Analytics

USERS

ID	First	Last
1	Jeffrey	Mackay

USER_SKILLS

UserID	Skill Name
1	SQL
1	JavaScript
1	NoSQL

USER_EXPERIENCE

ID	Role	Company
1	Solutions Architect	AppMax Inc
1	Solutions Engineer	Couchbase

Relational Tables

```
{
 "UserID": "1",

 "firstName":"Jeffrey",

 "lastName": "Mackay",

 "skills": ["SQL", "JavaScript", "NoSQL"]
 "experience": [
   {
    "role": "Solutions Architect",
    "company": "AppMax Inc",
   },
   {
    "role": "Solutions Engineer",
    "company": "Couchbase",
   }
  ]
}
```

JSON Document

Unlike relational databases, the application developer dictates the structure of the data

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SQL++ | Innovate, Iterate, and Scale apps faster

SQL-like queries for JSON

- Flexible data schema
- ANSI standard, including JOINS
- User Defined Functions
- Distributed SQL transactions

Query without compromise

- High-performance query engine
- Multidimensional scaling (MDS)
- Cost-based query optimizer
- Rich Indexing Options

Familiar tooling

- Create and test queries in the Query Workbench
- Visual Query Plan
- Index Advisor to optimize queries

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Couchbase SDKs

Ease of development

- Native support in programming languages
- Discoverable API via IDE
- Robust SDK supporting connection management, error handling, timeouts and retry

Feature rich

- Easily navigate Couchbase hierarchy
- Topology awareness
- Transactions support
- Sync and Async APIs
- Metrics & observability
- Security & encryption

Idiomatic

- Unified API across languages
- Consistent behavior
- Fluent and readable syntax
- Language specific design, when applicable.

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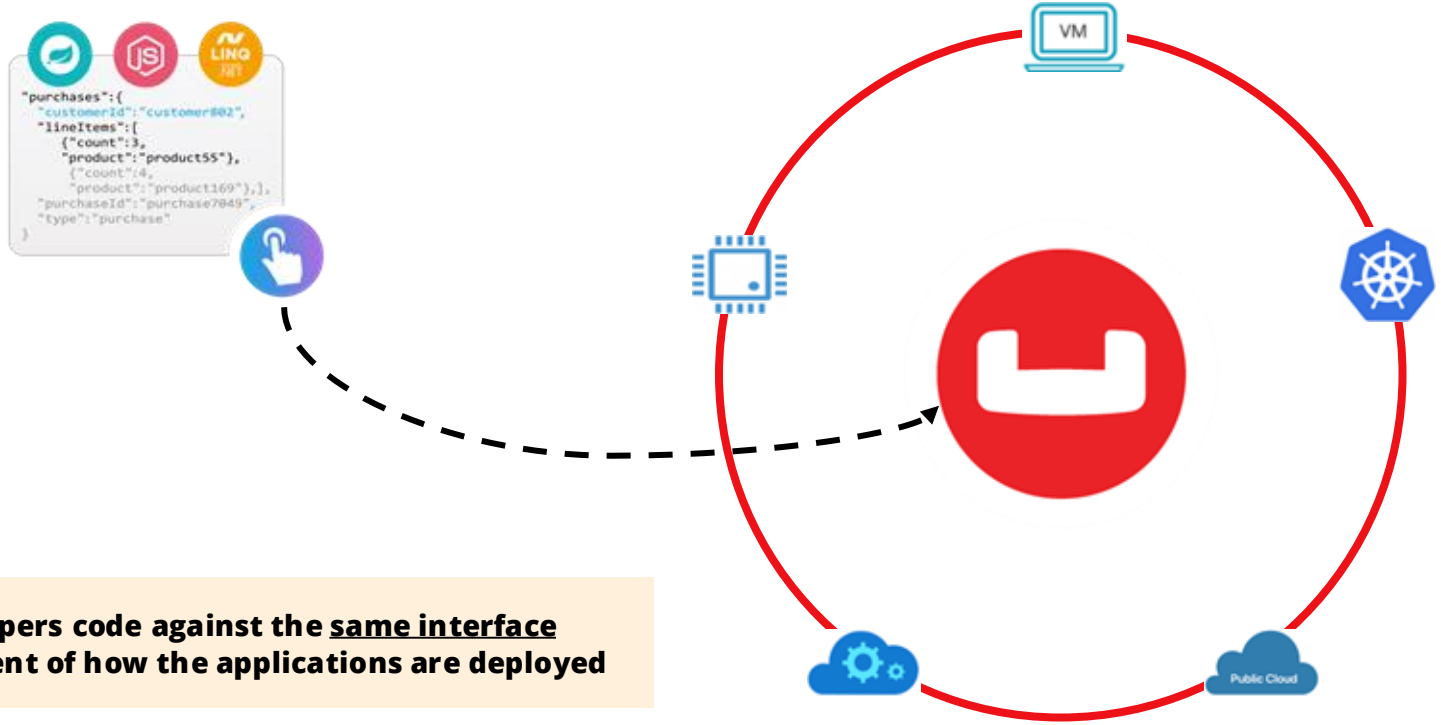
Consistent Development Experience on Any Infrastructure



Developers code against the <u>same interface</u> independent of how the applications are deployed

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Development | SDKs, Tools and Integrations

Couchbase SDKs

- 11 SDKs to choose from
- We strongly advise using the latest version of your preferred SDK

Couchbase Tools & Plugins

- VSCode plugin, Jetbrains IDE plugin, cbshell tool, Hashicorp Vault plugin, and Capella Terraform provider to streamline development and deployment

Connectors

- Kafka, Spark, ElasticSearch, BI connectors & tools to facilitate integration

REST

- Use Off the shelf HTTP clients with Couchbase RESTful interface for development and management

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Key Takeaways

Server Product Releases Highlights

July 2021

Couchbase Server **7.0**

- Scopes/Collections
- Transactions with SQL++
- Cost-Based Optimizer
- Index Advisor
- Prometheus Metrics
- Backup Service

May 2022

Couchbase Server **7.1**

- Magma Storage Engine
- UDFs with SQL++
- Analytics HA Support
- Tableau Connector
- Multi-Node Failovers
- System Event Log
- Multiple Certificates

May 2023

Couchbase Server **7.2**

- Time Series
- Analytics CBO
- Analytics Remote Links
- Change History Retention

March 2024

Couchbase Server **7.6**

- Vector Search
- Graph Traversals
- SSO/MFA
- Faster Rebalance
- Query without Indexes
- PowerBI Connector
- Eventing for Analytics

Q3 2025

Couchbase Server **8.0**

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8.0 Overview | Significant Enhancements Across All Services

- Vector Search that can **scale to billions of vectors** with high QPS and low TCO
- Query workload snapshots and reports to make it easier to **troubleshoot queries**
- Support **natural language** input and conversion to SQL++ queries

- Support bi-directional (Active-Active) **XDCR with mobile buckets**
- XDCR target clusters are aware of **incoming replications** for easier management
- **Log conflicts** when doc updated on both source/target clusters within conflict window

- **Native encryption-at-rest** with support for KMS integration for key management
- **Lower memory quota** for Magma buckets to 100MB with 128 vbuckets
- Allow a **new node to start serving** requests while the cache is being warmed up

Query & Index Services

Search Service

Eventing Service

XDCR Service

Data Service

Cluster Management

- Support text search based on **user-defined synonyms**
- Ability to **filter** which documents to index for search based on granular criteria
- Leverage the **partition replica** to improve the throughput of search queries

- Ability to **set options** like enable_curl or enable_debugger at the function-scope or bucket level
- Ability to **configure the number of nodes** on which an eventing function within a function scope executes

- **Auto-failover** for non-responsive disks and ephemeral buckets with no replicas
- **Add/remove** non-KV services on **existing nodes** without the need to add new nodes
- **Aggregate SDK client metrics** on server for easier monitoring and troubleshooting

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8.0 Overview | For Developers

Improve developer productivity with streamlined developer experience and support for AI use cases

AI-Enhanced Applications

Enable developers to use emerging AI technologies to enhance their adaptive apps

- Extend GSI to support vector indexes that can scale to billions of vectors with high QPS and low TCO
- Full integration of vector search APPROX_VECTOR_DISTANCE in SQL++ with efficient filtering
- Integrations with AI frameworks like LangChain, llamaIndex and Haystack

Developer Productivity

Make it easier for developers to get started and develop their first application

- Allow users not familiar with SQL++ to use natural language to query data
- Automatically refresh statistics to help the optimizer generate optimal query plans
- Support text search based on user-defined synonyms for more relevant search results

Query Performance Troubleshooting

Make it easier to identify performance issues for complex queries

- Query workload snapshots and reports to simplify query performance troubleshooting
- Reporting tool for users to generate summary report based on the collected stats
- Includes two time periods comparison to determine query performance changes

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8.0 Overview | For Ops Engineers

Empower Ops Engineers with heightened security and compliance, operational efficiency, and uptime assurance

Security & Compliance

Enable SREs to meet enterprise compliance and security requirements

- OOTB support for encryption-at-rest of sensitive information associated with the data service
- Support for key management with KMIP protocol based KMS
- Manually secure or restore local user accounts, and enforce password change to maintain strong security

Simple Management

Enable SREs to efficiently manage and support Couchbase clusters

- Aggregate SDK client metrics on server for easier end-end monitoring and troubleshooting
- Add or remove non-KV services of existing nodes without the need to add new nodes to cluster
- XDCR target clusters are aware of incoming replications for easier management

Uptime SLA

Support mission-critical deployments with stringent uptime SLA

- Auto-failover for non-responsive disks
- Allow auto-failover to occur even if there are ephemeral buckets with no replicas.
- Allow a new node to start serving requests while the cache is being warmed up in the background

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Thank You

Couchbase